BAU – FET – DCE. EE- BSE0401 MIDTERM EXAM 45 MIN

- SHOW YOUR COMLETE CALCULATIONS.

Q1- (15 points) Select the correct answer:

1-1) For an original (first) amount of JD 1500, at a Simple Interest Rate i=10% per year, the Interest Amount (I) after 6 years is:

(A) JD 900 B. JD 90 000 C. JD 2400

1-2) The missing Factor in the equation (----?---)(A/P,i.n) = (A/F,i.n) is:

A. (P/A,i.n) B. (F/P,i.n) (C) (P/F,i.n)

1-3) How long will it take to Withdraw JD 2160 from a deposit of JD 1000 at i=8% per year? (n= around? years)

A. 12.4 years (B) 10 years C. 8 years

Q2- (5points): For a project with a First cost = 4000, an Extra Cost at the end of the 4th year = 1300, and a S.V. = 500. If i=8% per year and n=7 years, then the Equivalent Single Value at the end of the 4th year is?

Eq. X (4) = ...+ 6343.1.....

Q3- (5 points): A project with a First Cost = 1000 and an Annual Operating cost for the first 4 years =1000, if i=8% per year and n=7 years, then the Equivalent Gradient Value (G) for the last 4 years is closest to?

Eq. G = ...+ 1168.19.....

Q4- (5points): A project with an Extra Cost, at the end of the 3rd year = 1200, and an annual Income = JD 1000, Decreased by 100 yearly over n=6 years. If i=8% per year, then the equivalent Annual Value is around:

Eq. A = ... 900 .....

BEST WISHES.

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erest Rate

i.n) is: $\frac{P}{F} \times \frac{A}{P} = \frac{A}{F}$

D 1000 / P

s,

Q.1

1) original amount = 1500 $i = 10\%$ $n = 6$

$I = P \cdot n \cdot i = 1500 \times 10\% \times 6 = 900$ TD

2) $\frac{\cdot P}{\cdot F} \times \frac{A}{P} = \frac{A}{F}$ $\boxed{\frac{P}{F}}$

3) $i = 8\%$ $P = 1000$ $F = 2160$ $n = ?$

$F = P(1+i)^n$

$2160 = 1000(1+8\%)^n$

$\frac{2160}{1000} = \frac{1000}{1000}(1.08)^n \Rightarrow \ln 2.16 = \ln(1.08)^n \Rightarrow n = \frac{\ln(2.16)}{\ln(1.08)} = 10$ year

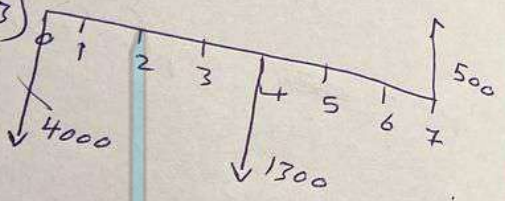
Q.2 First cost = 4000 extra cost at 4 = 1300 S.V = 500 $i = 8\%$ $n = 7$

$eq_A(4) = ?$

$eq_A(4) = 4000(F/P, 8\%, 4) + 1300 - 500(P/F, 8\%, 3)$

1.36 0.7938

$= +6343.1$ ↑

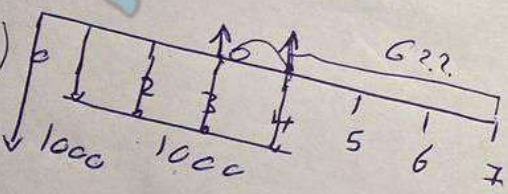


Q.3 $i = 8\%$ Get the last 4 Year ??

$eq_P6 = eq_A(7) = 1000(F/P, 8\%, 3) + 1000(F/A, 8\%, 4)(P/F, 8\%, 1)$

1.26 4.506 0.9259

$eq_P(6) = +5432.1054$ ↑



$G = P(G/P, 8\%, 4)$ $\frac{1}{\frac{P}{G}}$

$= 5432.1054(0.215)$

$G = +1167.9$ ↑

Q.4

$i = 8\%$

$A = 1000$ decreased. $G = 100$ eq A ??

$eq\,X(6) = 1200\,(P/F, 8\%, 13) - [P(A) - P(G)]$

0.7938

$eq\,X(6) = -2618.14$

$P(A) - P(G) = A(P/A, 8\%, 6) - G(P/G, 8\%, 6)$

$= 1000(4.623) - 100(10.523)$

$= 3570.7$

$A = P(A/P, 8\%, 6)$

$= -2618.14(.2163)$

$eq\,A = -566.3$

